MIT Mathematics ORP Lecture Notes Ezra Guerrero 02/02/24

The h-Cobordism Theorem

Goal: We will prove Smale's h-cobordism theorem, which says:

Let $n \geq 5$, and W^{n+1} a smooth, compact, oriented, simply connected h-cobordism from X^n to Y^n. Then, $W \cong_{\text{diffeo}} X \times [0,1]$.

Before proving the theorem, let's recall a few key facts:

- Every smooth, compact manifold pair $(M, \partial M)$ admits a handle decomposition, with handles attached in order of increasing index.
- Given a k-handle and a $k+1$-handle, they can be cancelled, provided the belt sphere of the k-handle and the attaching sphere of the $k+1$-handle intersect geometrically once.
- An h-cobordism W^{n+1} from X^n to Y^n consists of a smooth $n+1$-manifold W with boundary $\partial W = -X \sqcup Y$ such that the inclusion maps $i : X, Y \hookrightarrow W$ are homotopy equivalences.

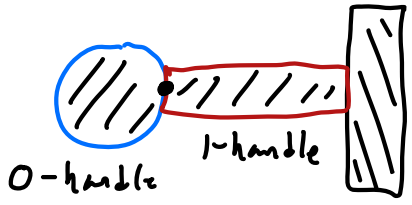
The idea of the proof is to cancel all handles of a handle decomposition of W. Since such a handle decomposition consists of attaching nothing to $X \times [0,1]$, it will follow that $W \cong_{\text{diffeo}} X \times [0,1]$ as desired.

We cancel our handles in three steps:

1) Cancel all 0- and $n+1$-handles
2) Trade all 1-handles for 3-handles
   (dualization means we also have no n-handles)
3) Remove all remaining handles via the Whitney Trick.

Step 1

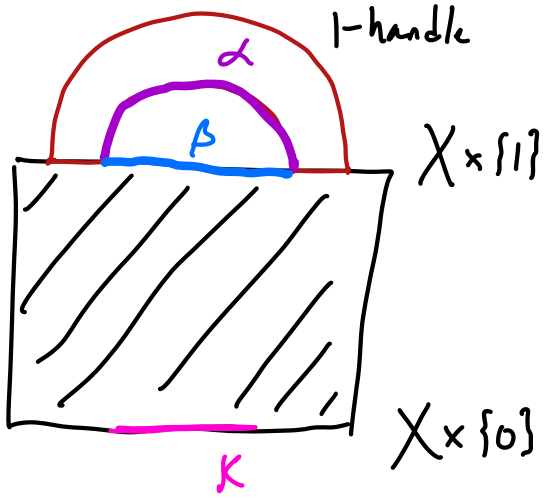
Because of dualization, it suffices to show we can remove all 0-handles. Since 0-handles are attached along their empty attaching region and W is connected, there must be a 0-handle connected to $X \times [0,1]$ via a 1-handle, as no other handle has disconnected attaching region. Evidently, the belt sphere of the 0-handle ($\cong S^n$) and the attaching sphere of the 1-handle ($\cong S^0$) must intersect exactly once, so these handles can be cancelled. Repeating this argument until there are no 0-handles left we may assume there are no 0-handles left in the decomposition.



Step 2

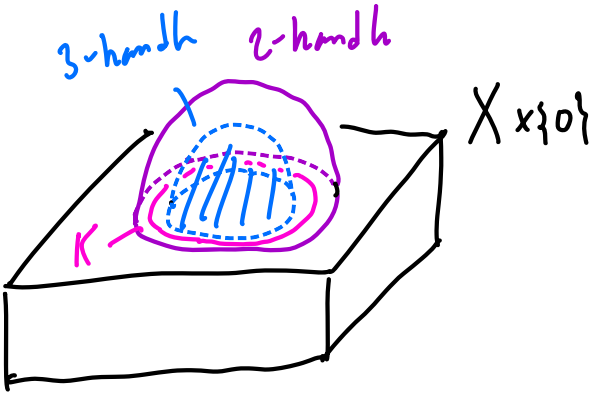
Let W_2 be the union of $X \times [0,1]$ and all 1- and 2-handles of W. Let M_2 be the new boundary, so that

$$\partial W_2 = -X \sqcup M_2.$$



We will introduce a 2,3 cancelling handle pair in such a way that the 2-handle cancels with a 1-handle and repeat this process until we have traded all 1-handles away for 3-handles.

Consider an embedded disk in X with boundary K. Thickening up this disk we produce our cancelling 2,3 pair. To cancel the 1-handle, consider an arc α in M_2 parallel to the core of the 1-handle. Attach to α an arc β in $X \times \{1\}$ with the same endpoints as α. This gives a circle $\alpha \cup \beta$ in M_2 along which we can attach a 2-handle. This 2-handle would cancel the 1-handle by construction. Hence, we just have to prove we can do both of them at once.

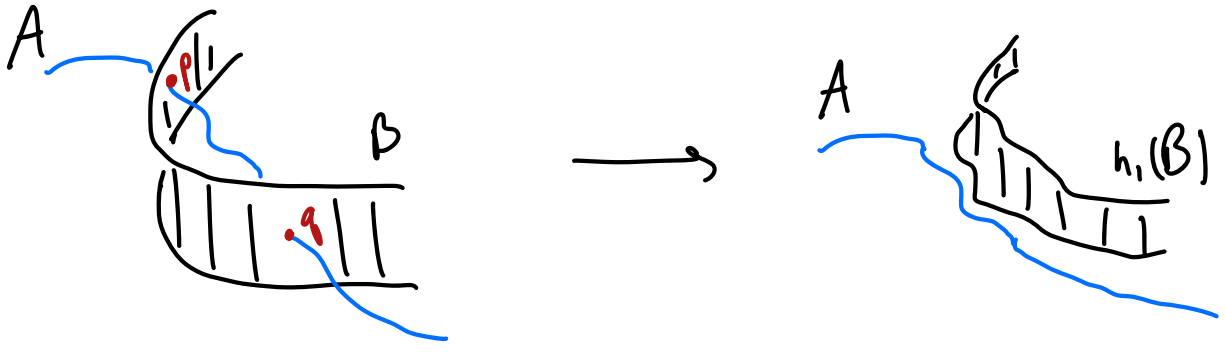


For this, it suffices to show we can isotope the respective attaching spheres of the 2-handles to each other. Since $\dim M_2 \geq 4$, it actually suffices to show they are homotopic.

For this, recall that $\pi_1(W) = 1$. Since the attaching region of a k-handle is simply connected for $k \geq 3$, all generators of π_2 must be killed by the time we have added our 2-handles, so $\pi_1(W_2) = 1$. Thus, $\alpha \cup \beta$ and K are homotopic in W_2. In order to push this homotopy into M_2, we push off of $X \times [0,1)$ and radially away from the cores of our handles into M_2. We can do this because of dimension reasons we may assume our homotopy is disjoint of all the cores.

Step 3

Finally, we will cancel all other handles in pairs. We can show that we can pair our remaining handles such that in each pair, their indices are consecutive and their algebraic intersection is 1. If we could arrange that their geometric intersection was 1, we would be able to cancel them, which after repeated application completes the proof. To do this, we use the Whitney Trick:



The Whitney Trick

Let A and B be two submanifolds of V of complementary dimension such that $\dim V \geq 5$, $\dim A \geq 3$ and V is simply connected. If A, B intersect transversely and p, q are two consecutive opposite sign intersection points, then we can find an isotopy $h_t : V \to V$ such that h_t is the identity away from A and B and

$$h_1(A) \cap B = A \cap B - \{p, q\}.$$

We have set things up so the conditions of the theorem apply above, so we can complete Step 3 and the proof. ∎